

Scalable String Reconciliation By Recursive Content-Dependent Shingling (RCDS)

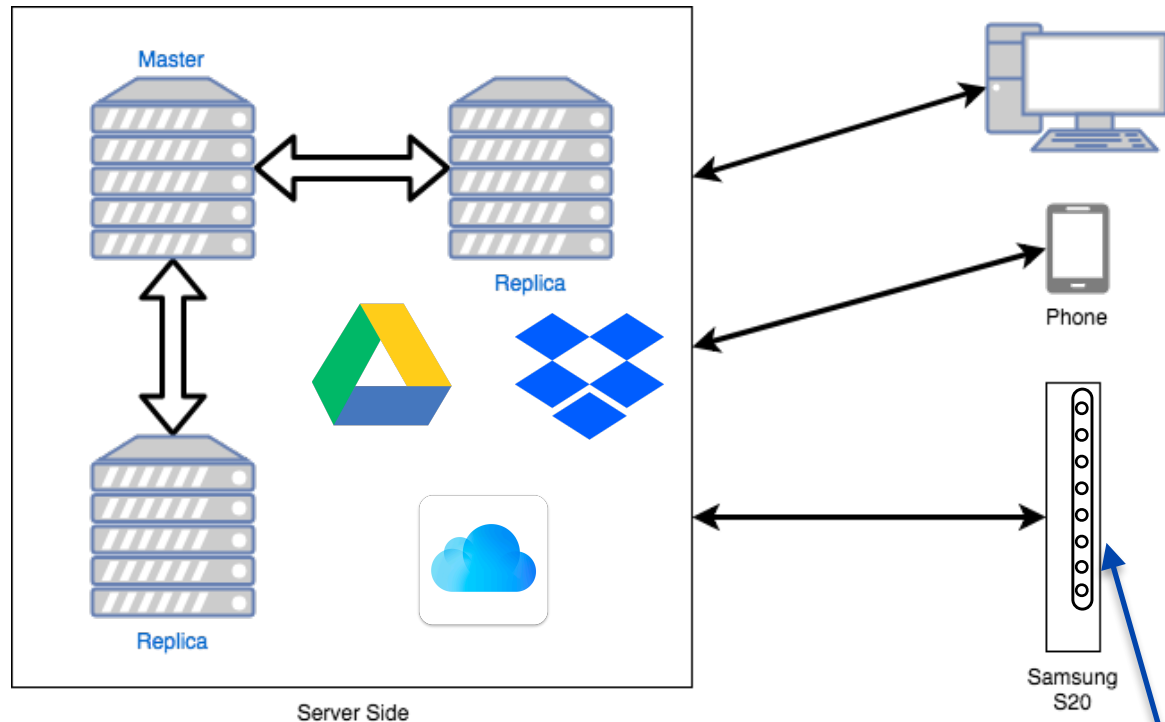
Bowen Song
Prof. Ari Trachtenberg

sbowen, trachten@bu.edu



Motivation - Distributed File System Clients

(Li, Z., ToMPECS, 2019)



Presentation Outline

1. Introduction

1. Problem Definition
2. Applications
3. State of the Art
4. Our Contribution

2. Main Protocol

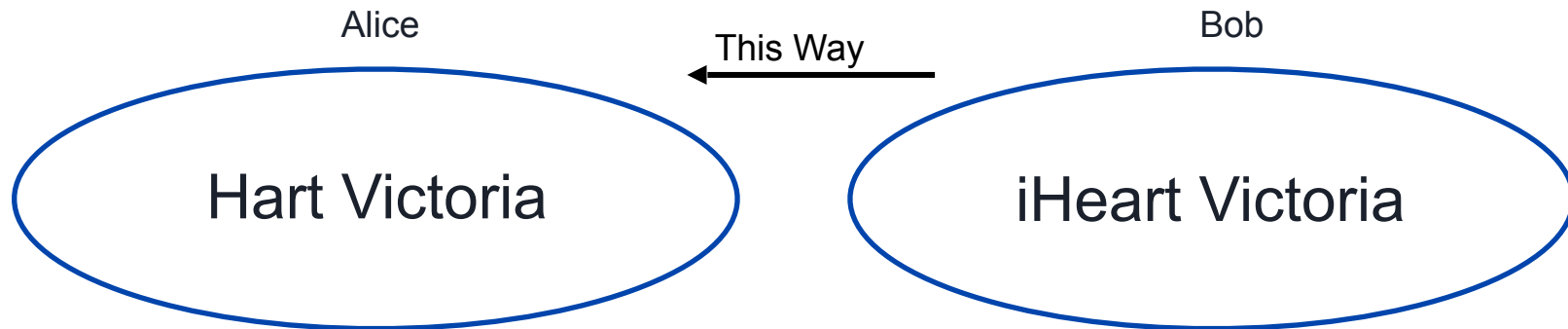
1. Analysis & Failure Modes
2. Experiments & Evaluations

3. Conclusion & Future Work



The Problem of String Reconciliation

1. Minimize the total amount of communication
2. Determine string difference



The Problem of String Reconciliation

1. Minimize the total amount of communication
2. Determine string difference

Position Reference & Data

Applications

1. Gossip Protocol: Asynchronous Edits
2. Distributed File Systems: Client-Sync + Internal Data Migration & Sync
3. Content-Delivery Networks: Parallel Download
4. Other Use Cases: Symbol (DNA) Sequencing

Existing Work

1. Matching Partitioned Data

- (Tridgell, A., 1999) (Muthitacharoen, A., SIGOPS 2001) (Yan, H., INFOCOM 2008)

2. Reduce to Set Reconciliation

- (Agarwal, S., TPDS 2006) (Trachtenberg, A., US Patent 2014) (Mitzenmacher, M., et al., arXiv 2018)

3. Coding Theory

- (Levenshtein, V. I. Soviet Physics 1965) (Orlitsky, A., IEEE Int. Symp. 2002) (Sala, F., IEEE Trans 2016)

4. Type Specific String Reconciliation

- (B. Saha, et al., FOCS 2016)

Recursive Content-Dependent Shingling (RCDS)

1. General Approach

1. Reduce to a multi-set of substrings
2. Use set reconciliation primitives
3. Unambiguously reconstruct the string from the set

2. Scalability

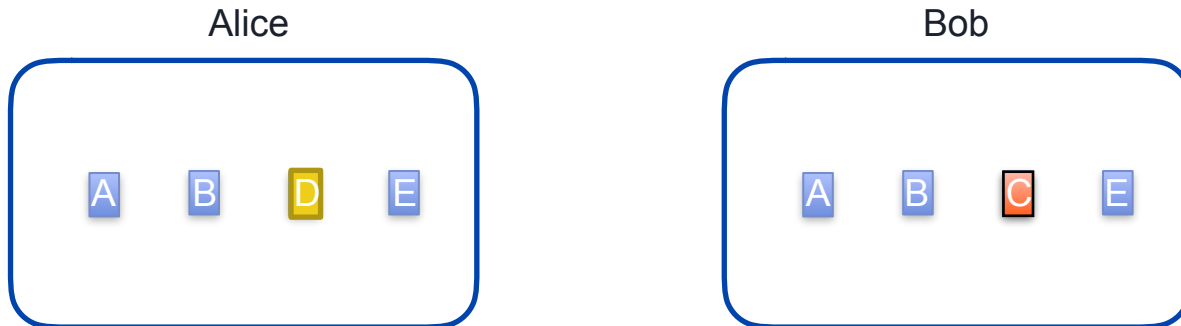
1. Sub-linear communication complexity (empirical)
2. Computationally scalable to long strings

The Problem of Set Reconciliation

1. Determine the union of the two sets
2. Transfer their symmetric differences
3. Information-Theoretic Lower Bound

Minsky, Y., Allerton, 2002

Goodrich, M.T., Allerton, 2011

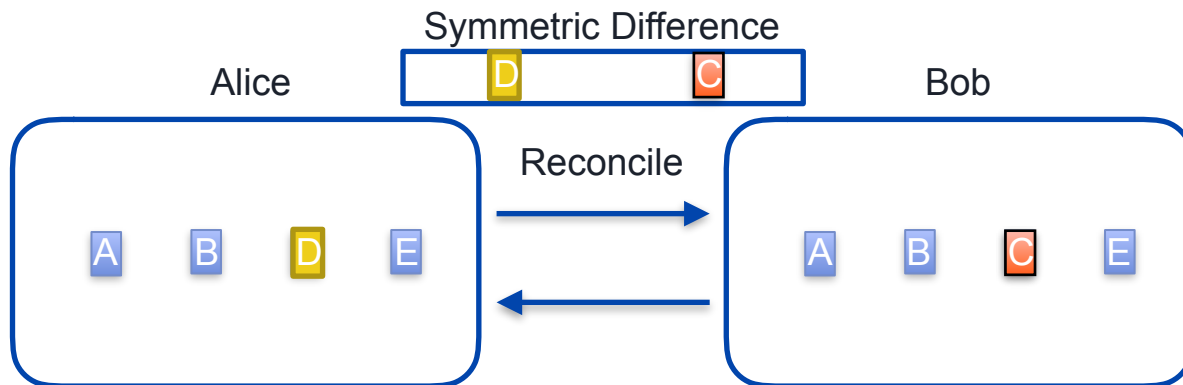


The Problem of Set Reconciliation

1. Determine the union of the two sets
2. Transfer their symmetric differences
3. Information-Theoretic Lower Bound

Minsky, Y., Allerton, 2002

Goodrich, M.T., Allerton, 2011

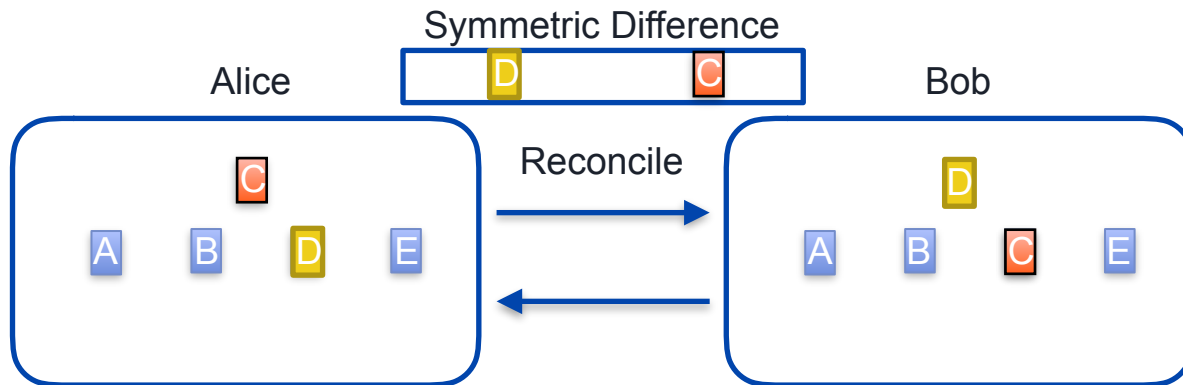


The Problem of Set Reconciliation

1. Determine the union of the two sets
2. Transfer their symmetric differences
3. Information-Theoretic Lower Bound

Minsky, Y., Allerton, 2002

Goodrich, M.T., Allerton, 2011



Content-Dependent Partition - Rolling Hash

Rolling Window Size: $w = 3$

$$\Sigma^w \gg s$$

Content Hash Space: $s = 16$

$$\Sigma = |\text{string content alphabet}|$$

i	H	e	a	r	t		V	i	c	t	o	r	i	a
---	---	---	---	---	---	--	---	---	---	---	---	---	---	---

Content-Dependent Partition - Rolling Hash

Rolling Window Size: $w = 3$

$$\Sigma^w \gg s$$

Content Hash Space: $s = 16$

$$\Sigma = |\text{string content alphabet}|$$

i	H	e	a	r	t		V	i	c	t	o	r	i	a
---	---	---	---	---	---	--	---	---	---	---	---	---	---	---

--	--	--	--	--	--	--	--	--	--	--	--	--

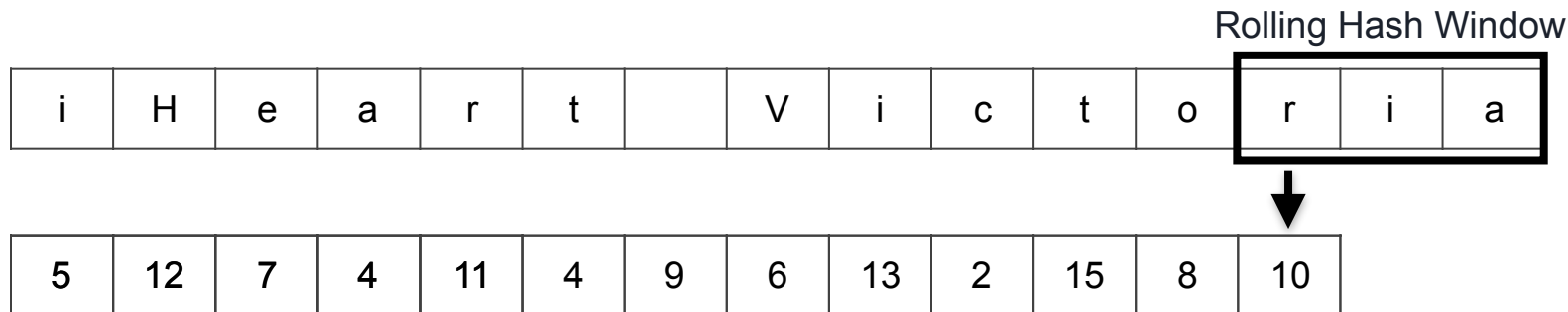
Content-Dependent Partition - Rolling Hash

Rolling Window Size: $w = 3$

$$\Sigma^w \gg s$$

Content Hash Space: $s = 16$

$$\Sigma = |\text{string content alphabet}|$$



Content-Dependent Partition - Local Minimum Chunking

Rolling Window Size: $w = 3$

Inter-Partition Distance: $h = 2$

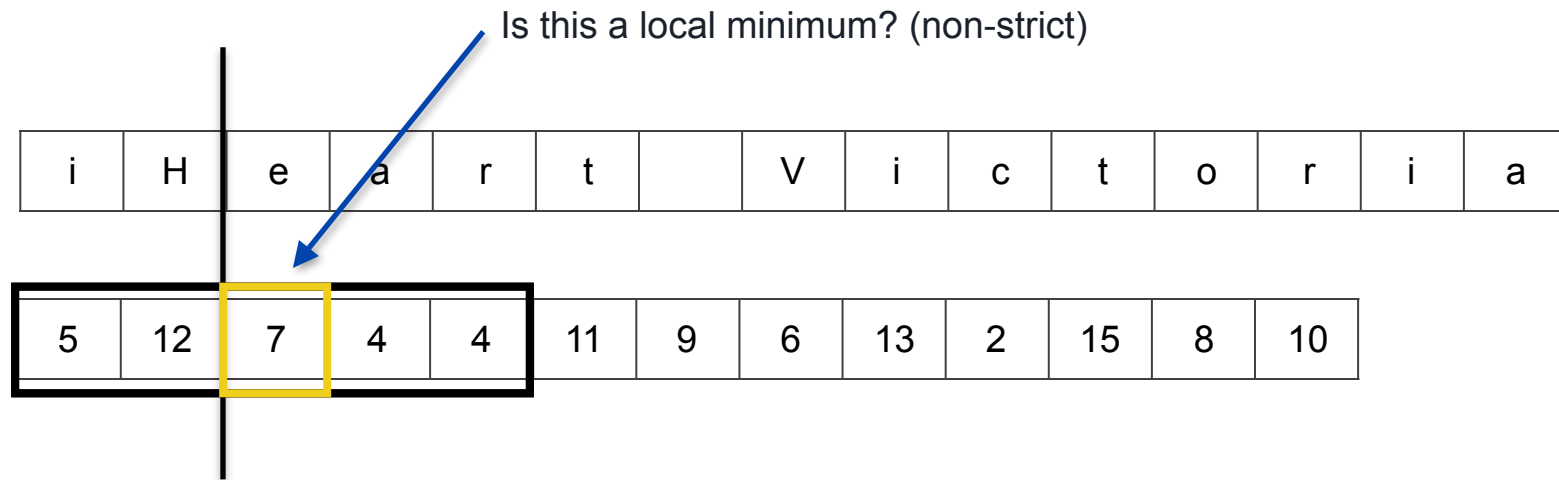
i	H	e	a	r	t		V	i	c	t	o	r	i	a
---	---	---	---	---	---	--	---	---	---	---	---	---	---	---

5	12	7	4	4	11	9	6	13	2	15	8	10
---	----	---	---	---	----	---	---	----	---	----	---	----

Content-Dependent Partition - Local Minimum Chunking

Rolling Window Size: $w = 3$

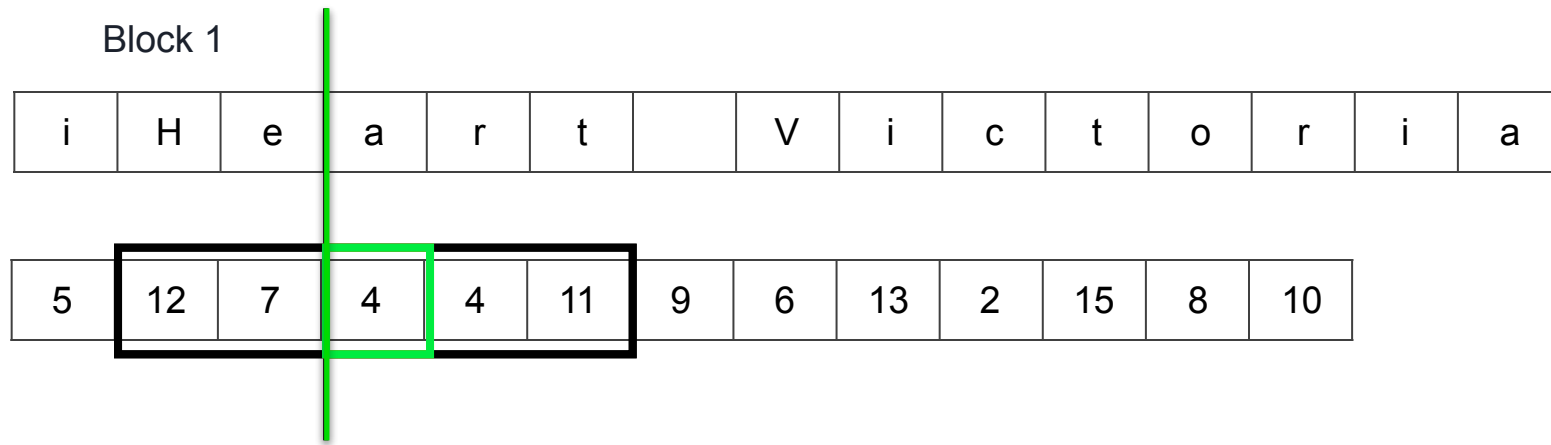
Inter-Partition Distance: $h = 2$



Content-Dependent Partition - Local Minimum Chunking

Rolling Window Size: $w = 3$

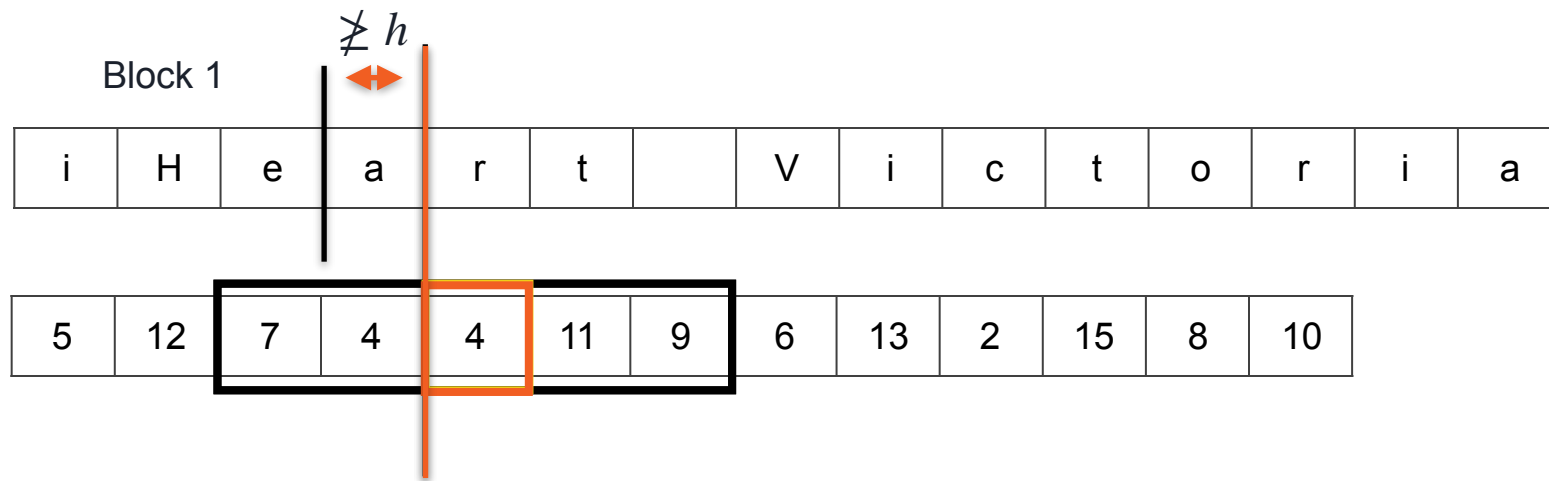
Inter-Partition Distance: $h = 2$



Content-Dependent Partition - Local Minimum Chunking

Rolling Window Size: $w = 3$

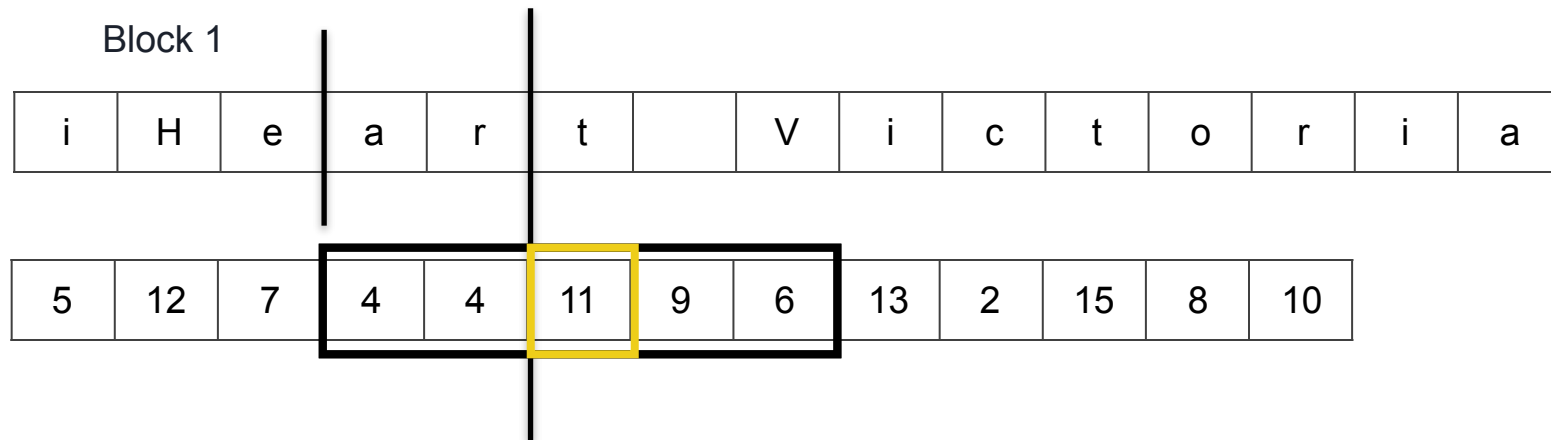
Inter-Partition Distance: $h = 2$



Content-Dependent Partition - Local Minimum Chunking

Rolling Window Size: $w = 3$

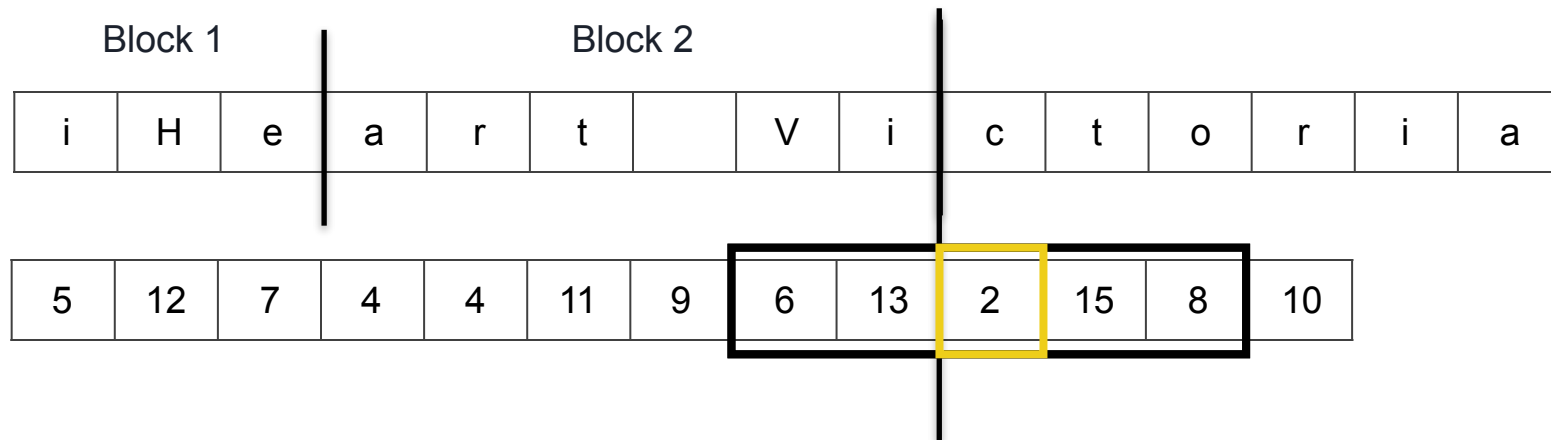
Inter-Partition Distance: $h = 2$



Content-Dependent Partition - Local Minimum Chunking

Rolling Window Size: $w = 3$

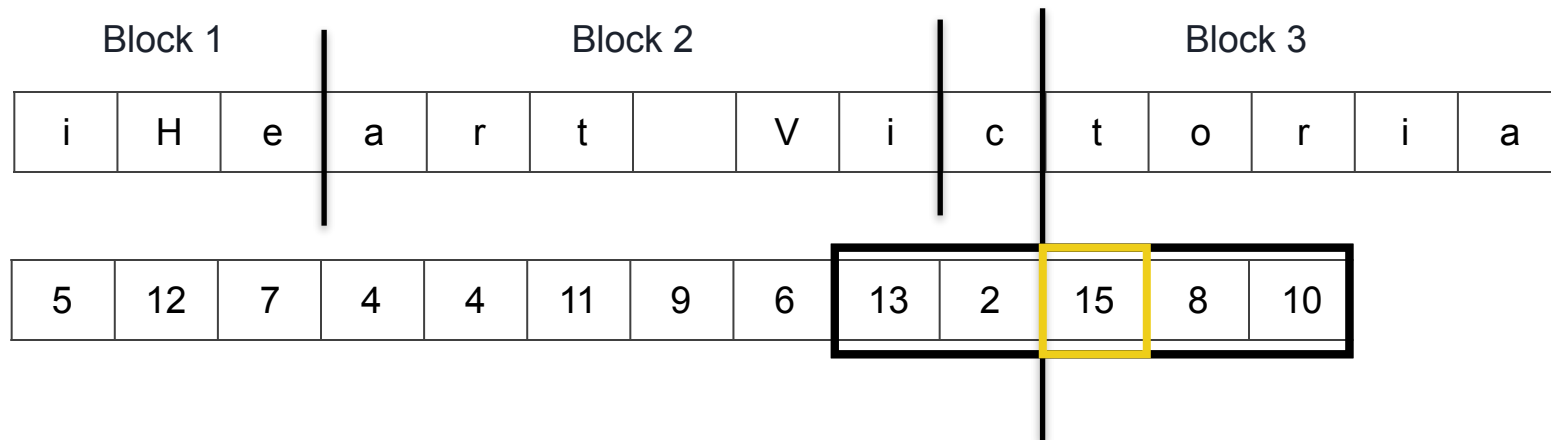
Inter-Partition Distance: $h = 2$



Content-Dependent Partition - Local Minimum Chunking

Rolling Window Size: $w = 3$

Inter-Partition Distance: $h = 2$



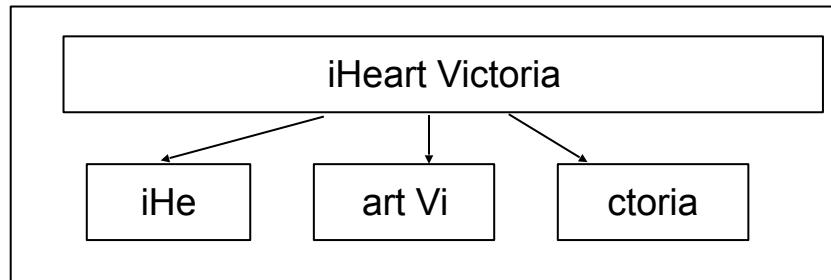
Content-Dependent Partition - Ideal Case

Block 1			Block 2						Block 3					
i	H	e	a	r	t		V	i	c	t	o	r	i	a
5	12	7	4	4	11	9	6	13	2	15	8	10		

Content-Dependent Partition - Ideal Case

Block 1			Block 2						Block 3					
i	H	e	a	r	t		V	i	c	t	o	r	i	a
5	12	7	4	4	11	9	6	13	2	15	8	10		

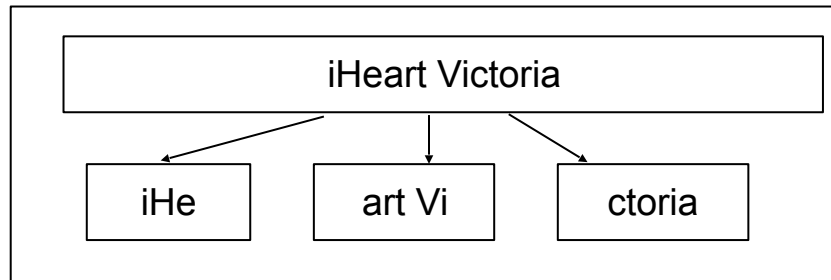
Alice



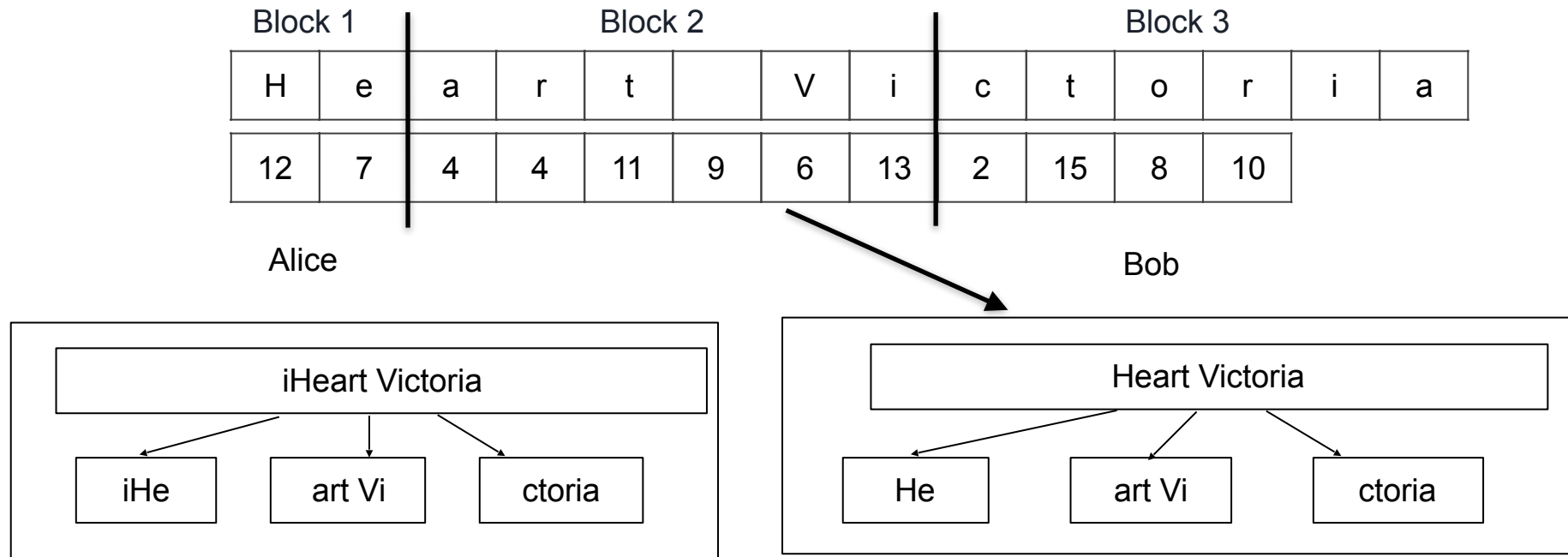
Content-Dependent Partition - Ideal Case

Block 1		Block 2						Block 3					
H	e	a	r	t		V	i	c	t	o	r	i	a
12	7	4	4	11	9	6	13	2	15	8	10		

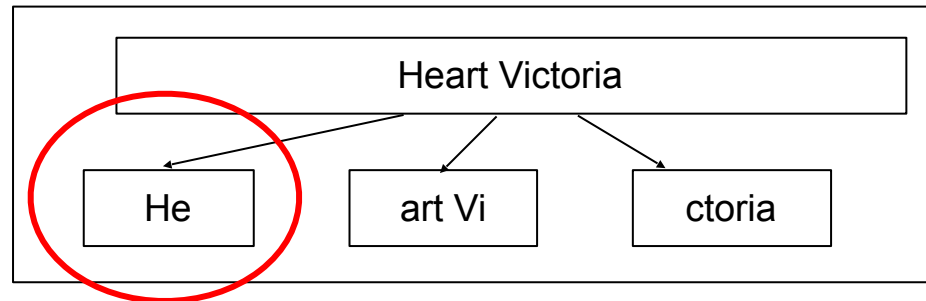
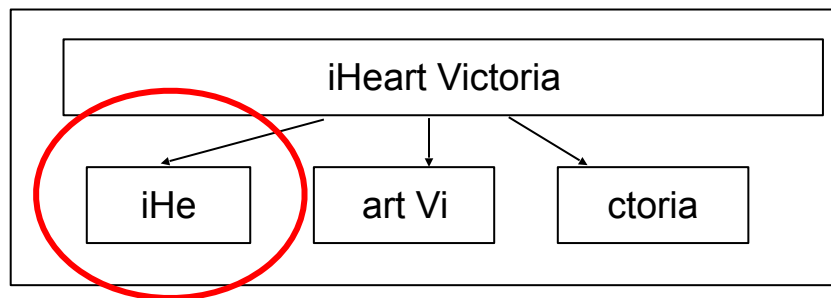
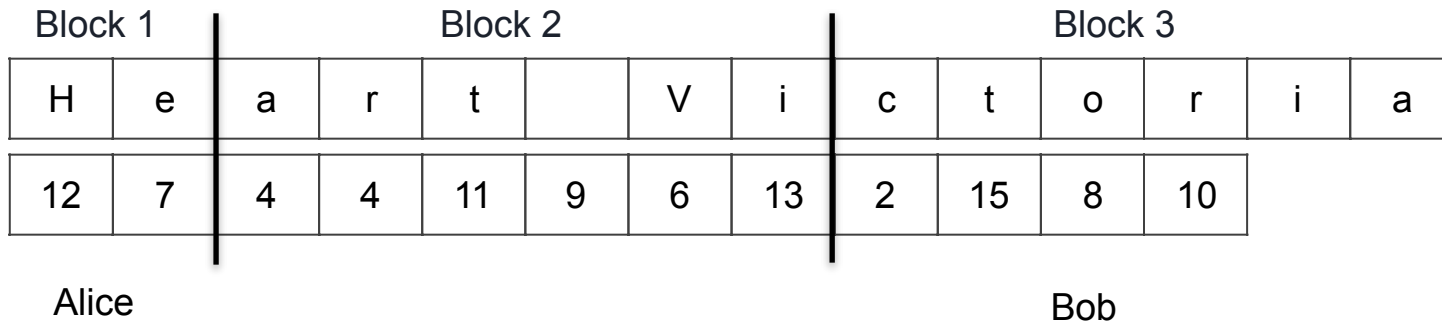
Alice



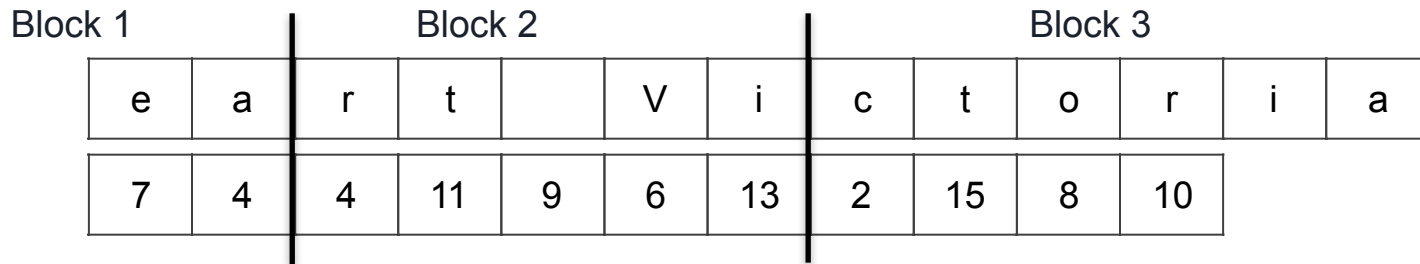
Content-Dependent Partition - Ideal Case



Content-Dependent Partition - Ideal Case

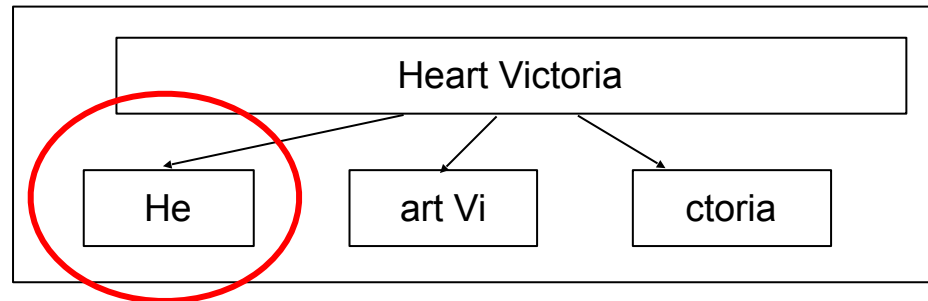
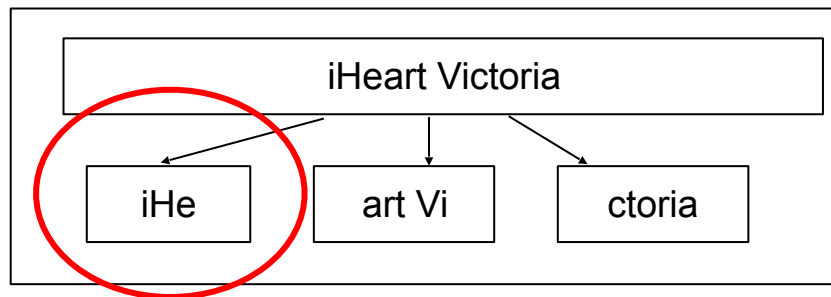


Content-Dependent Partition - Ideal Case



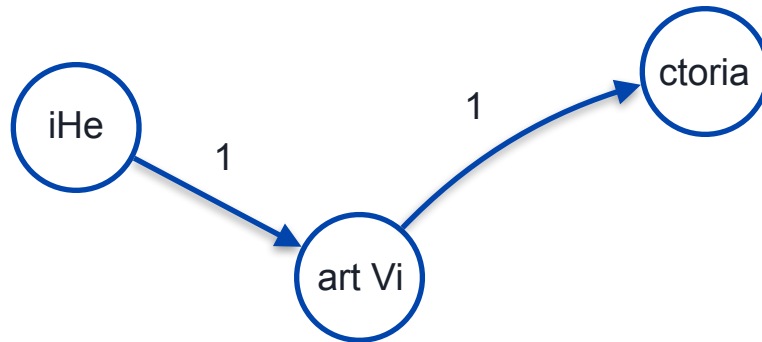
Alice

Bob



De Bruijn Digraph + Shingling + Backtrack

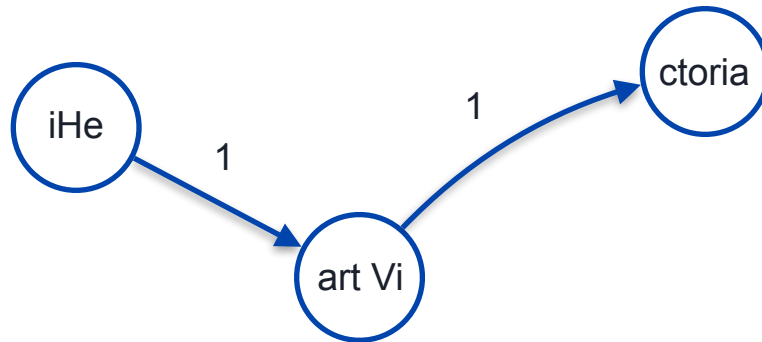
1. Brute Force
2. Exponential Time
3. Deterministic



iHeart Victoria

De Bruijn Digraph + Shingling + Backtrack

1. Brute Force
2. Exponential Time
3. Deterministic



iHeart Victoria

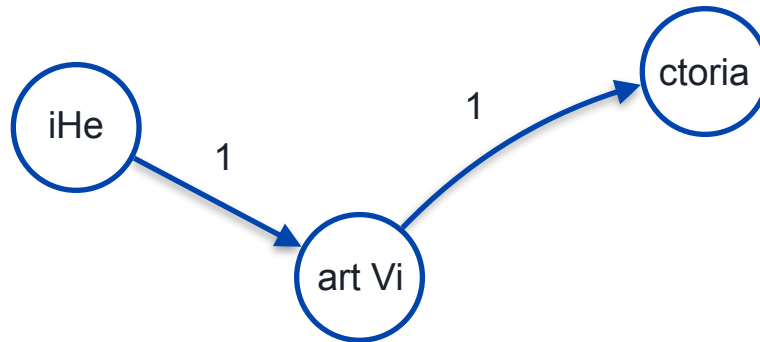
“.iHe”:1

“iHe”:“art Vi”:1

“art Vi”:“ctoria”:1

De Bruijn Digraph + Shingling + Backtrack

1. Brute Force
2. Exponential Time
3. Deterministic



iHeart Victoria

"": "iHe": 1

"iHe": "art Vi": 1

"art Vi": "ctoria": 1

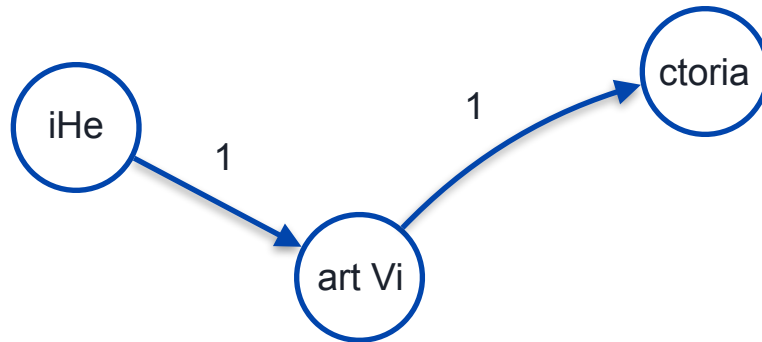
0: H("iHe"): 1

H("iHe"): H("art Vi"): 1

H("art Vi"): H("ctoria"): 1

De Bruijn Digraph + Shingling + Backtrack

1. Brute Force
2. Exponential Time
3. Deterministic



iHeart Victoria

0:H("iHe"):1

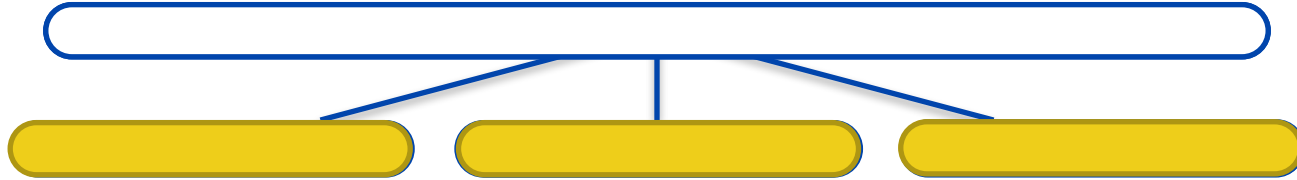
H("iHe"):H("art Vi"):1

H("art Vi"):H("ctoria"):1

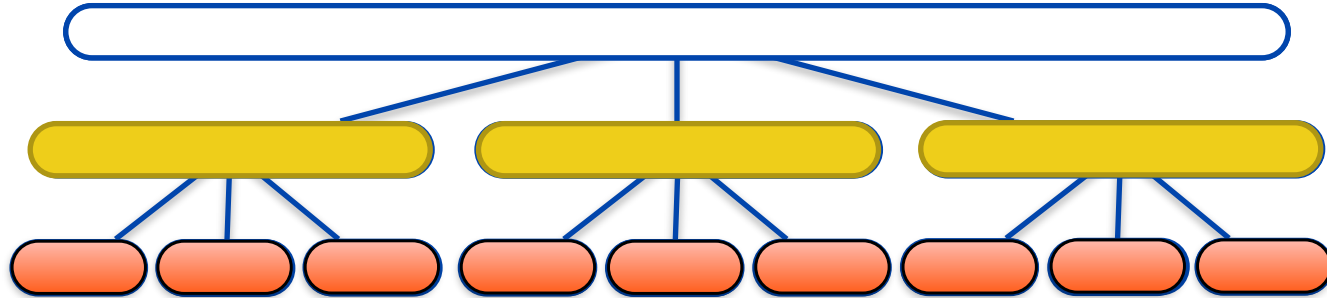
String to Set - $p=3$, $L = 3$



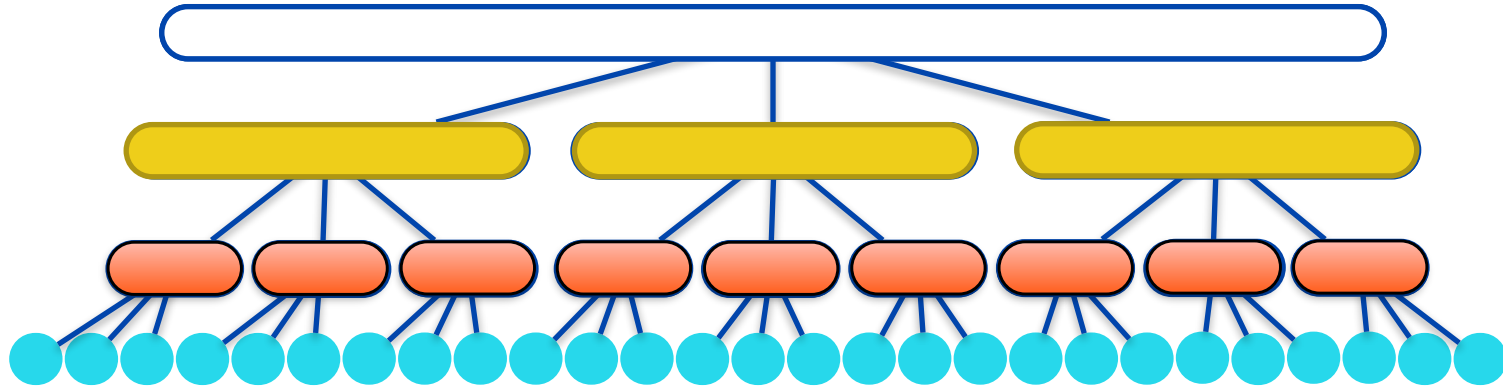
String to Set - $p=3$, $L = 3$



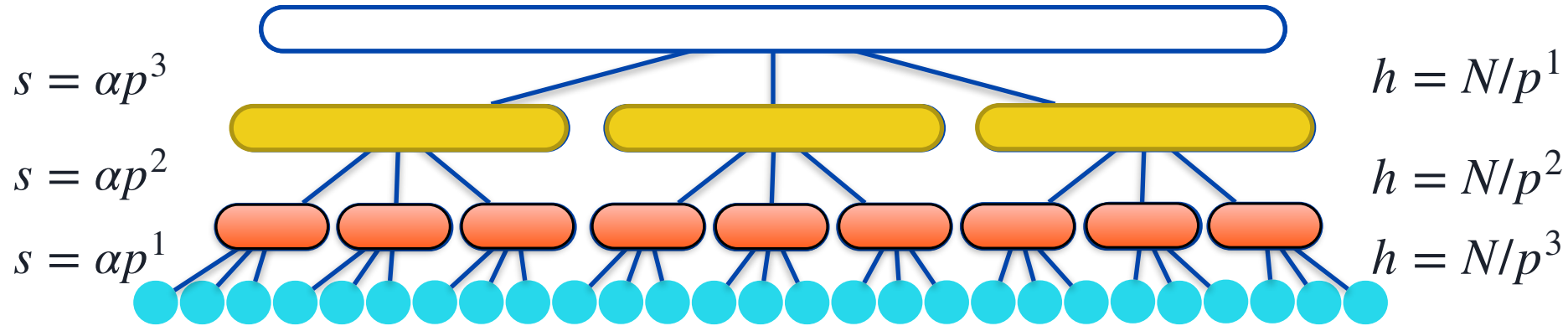
String to Set - $p=3$, $L = 3$



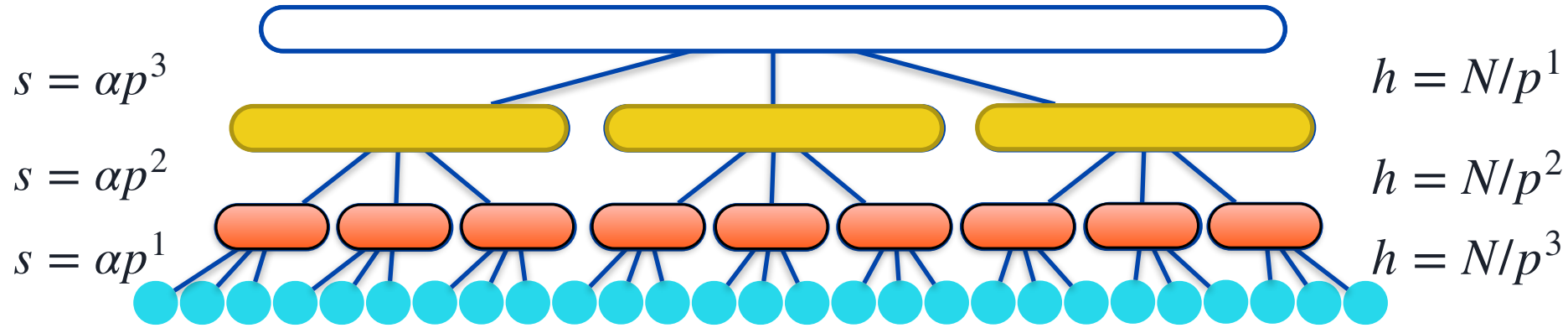
String to Set - $p=3$, $L=3$



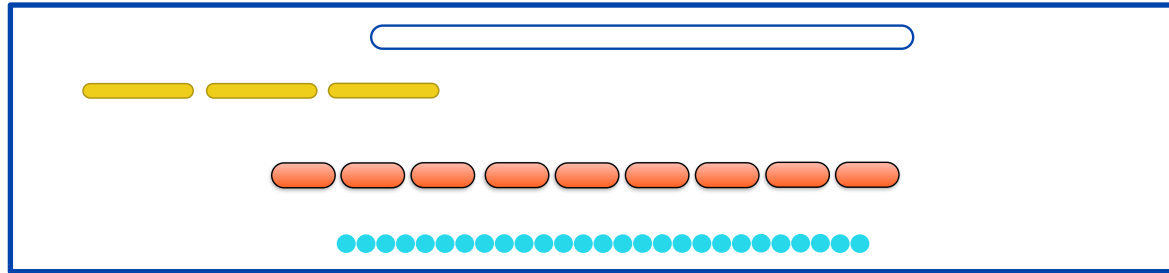
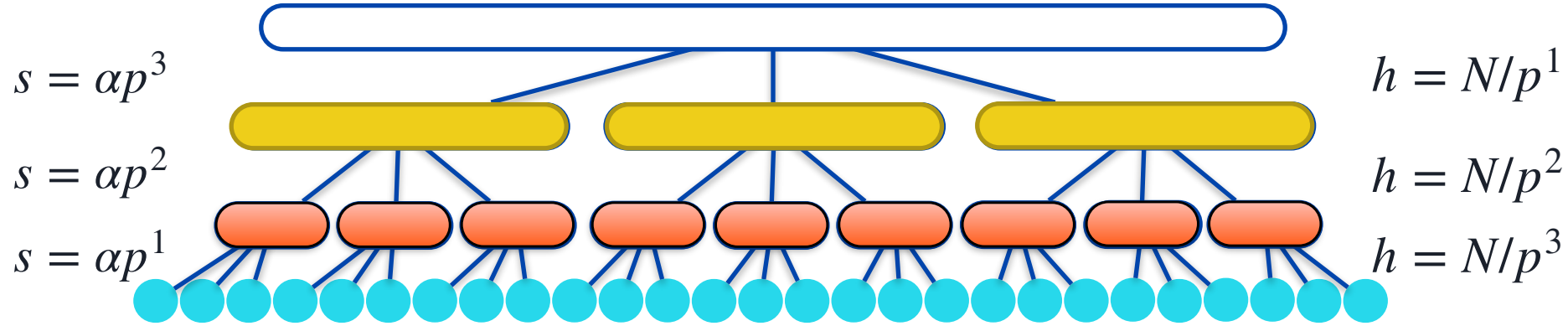
String to Set - $p=3, L=3$



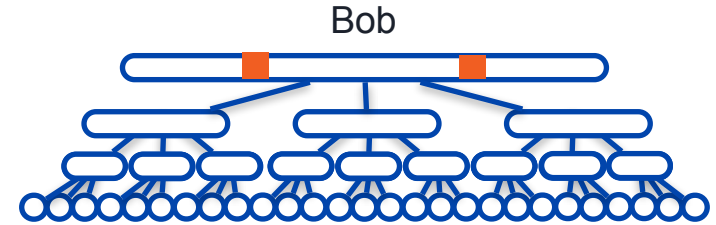
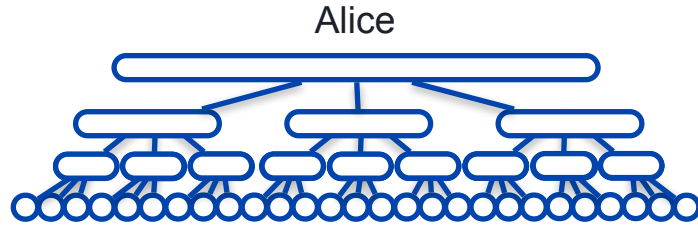
String to Set - $p=3, L=3$



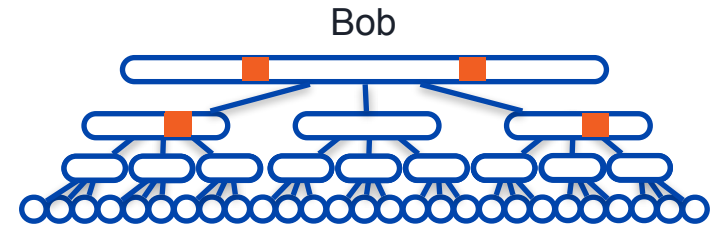
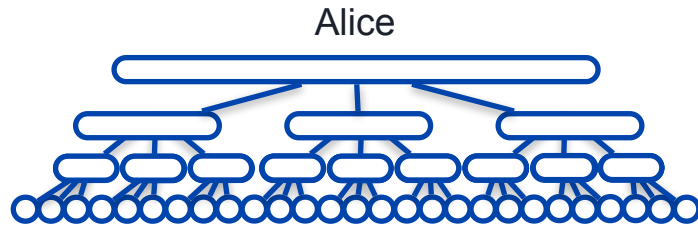
String to Set - $p=3, L=3$



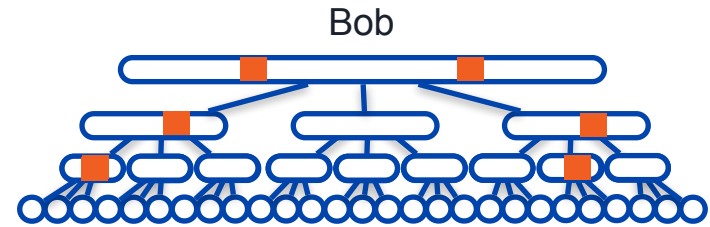
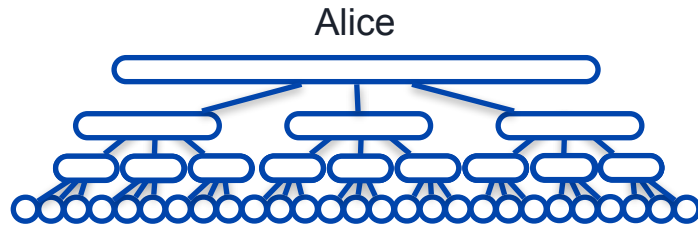
Main Protocol



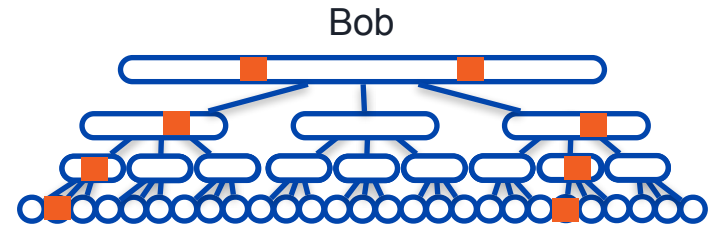
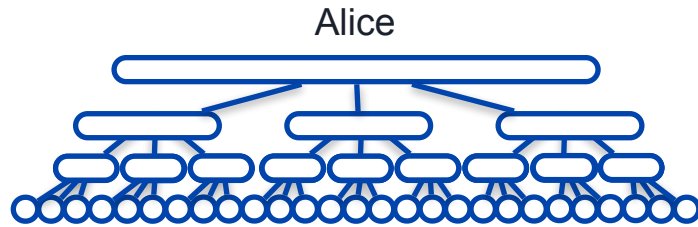
Main Protocol



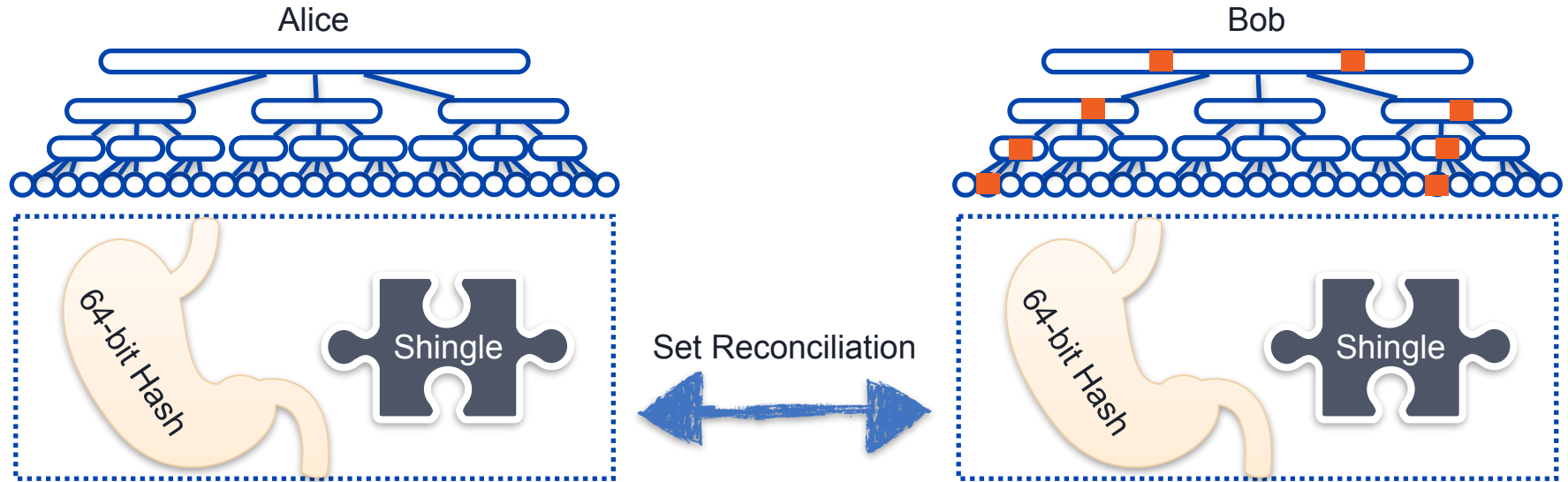
Main Protocol



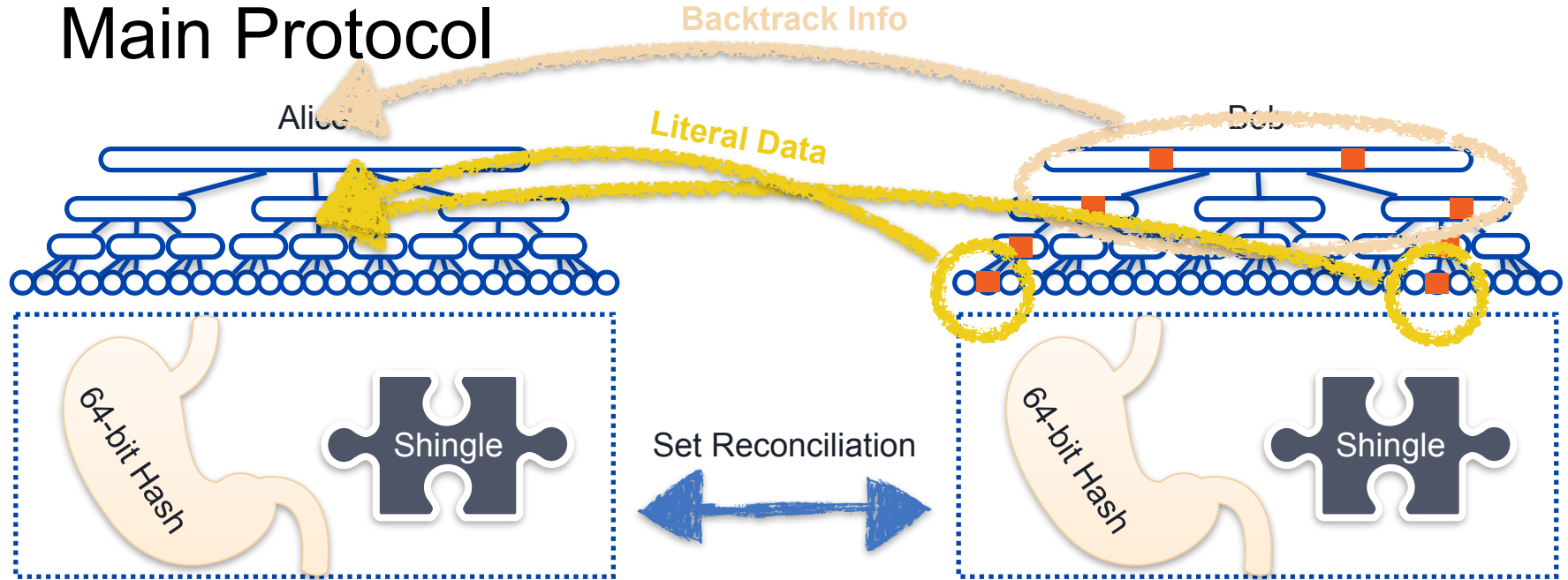
Main Protocol



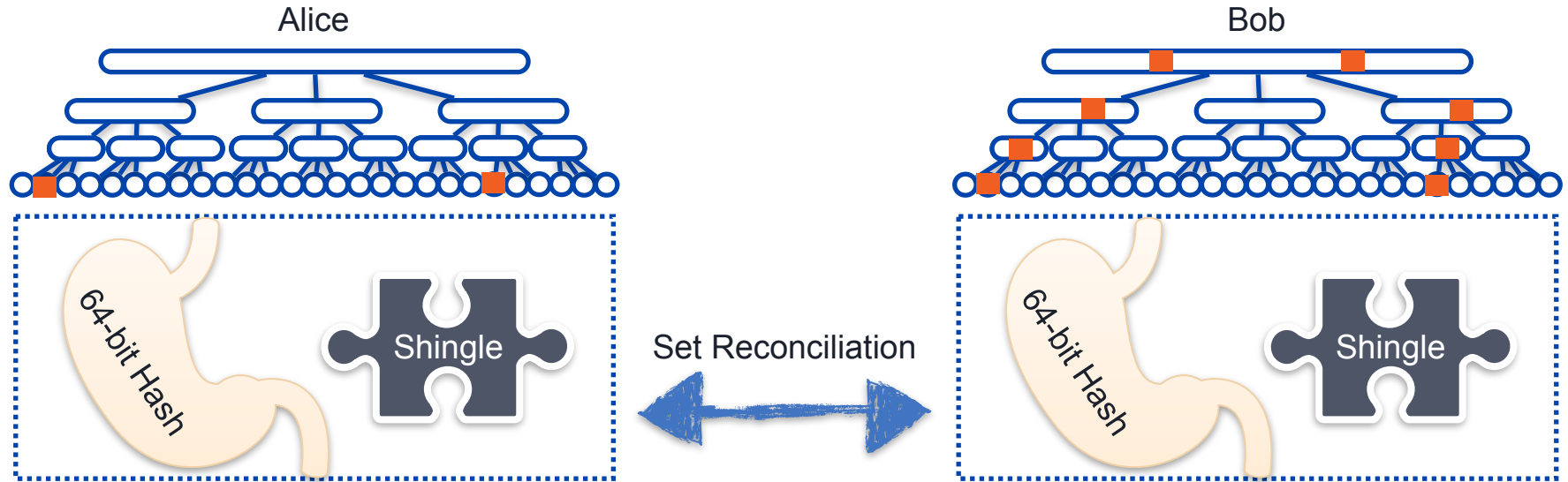
Main Protocol



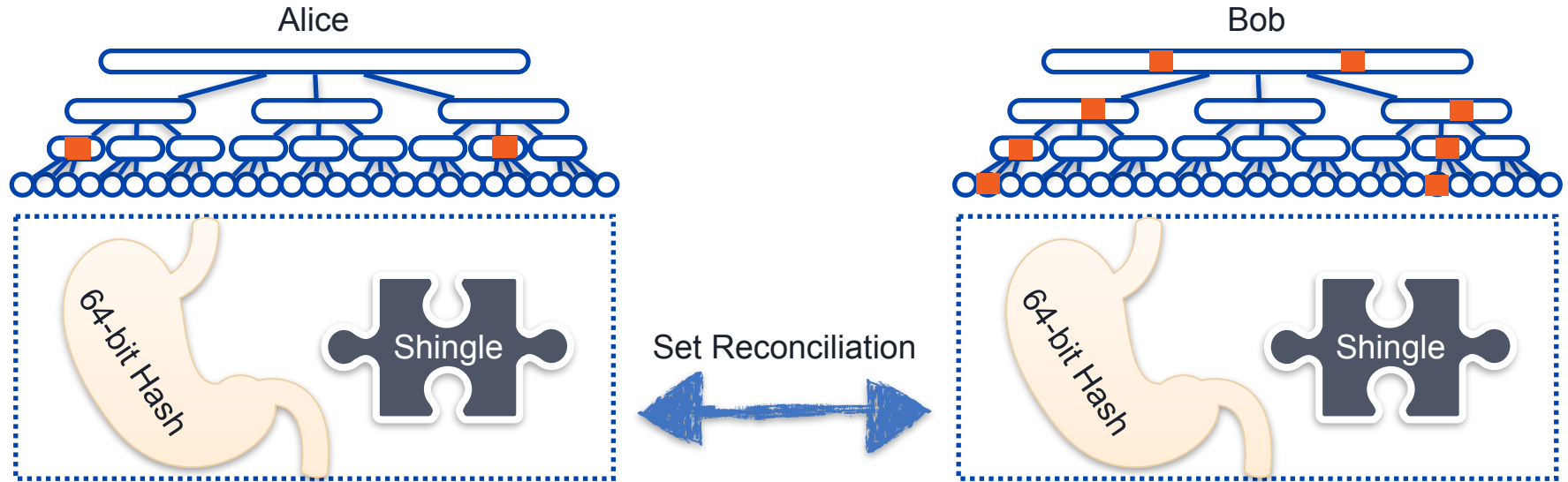
Main Protocol



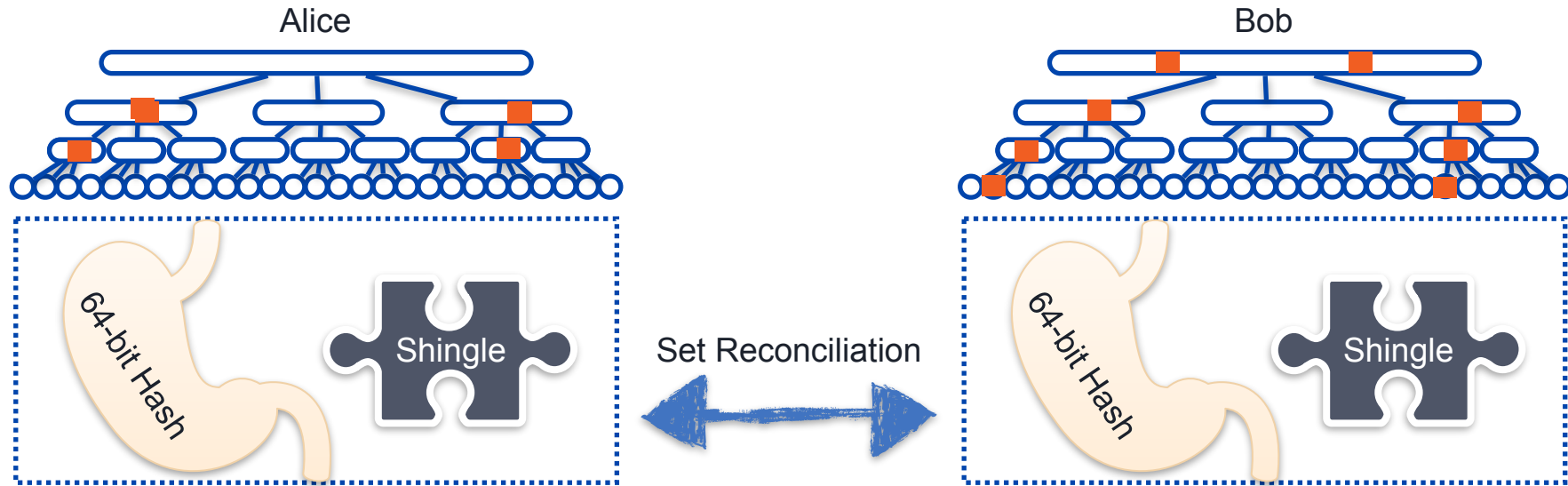
Main Protocol



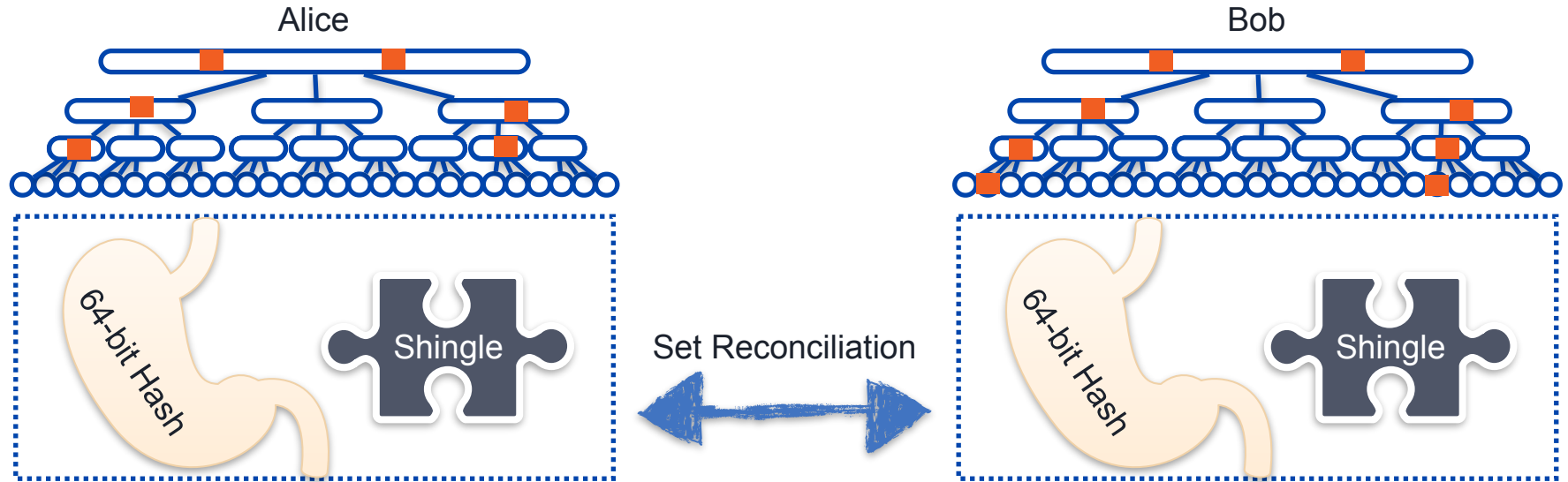
Main Protocol



Main Protocol



Main Protocol



RCDS Complexity

Time: $O(N \lg(N)L + dL\gamma) + \text{set reconciliation time}$

Communication: $O(dLb + dT)$

Space: $O(p^L + dT) + \text{set reconciliation space}$

p = Max # of partitions

L = Levels of recursion

T = Average terminal string length

d = Edit distance

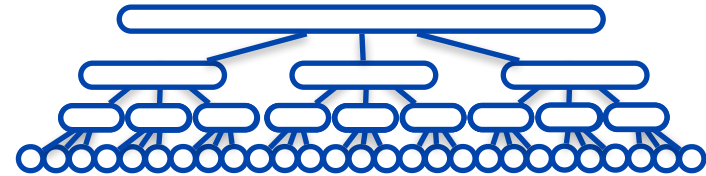
N = Input string length

γ = Backtracking time

b = # of bit per hashed shingle

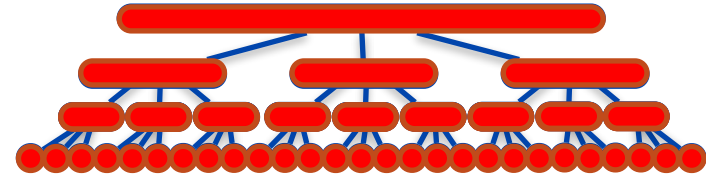
Worst-Case

1. Edit Locality
2. Cascading Mismatch
3. Duplicated Partitions



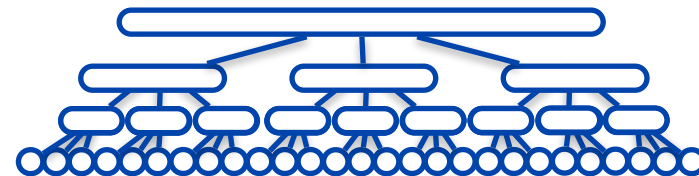
Worst-Case

1. Edit Locality
2. Cascading Mismatch
3. Duplicated Partitions



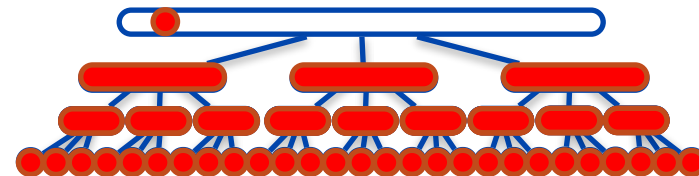
Worst-Case

1. Edit Locality
2. Cascading Mismatch
3. Duplicated Partitions



Worst-Case

1. Edit Locality
2. Cascading Mismatch
3. Duplicated Partitions

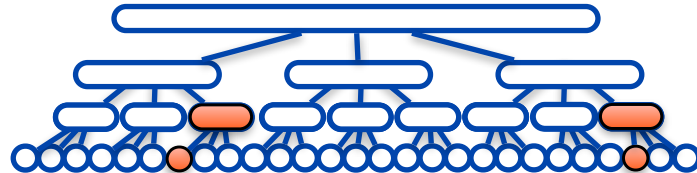


Worst-Case

1. Edit Locality
2. Cascading Mismatch
3. Duplicated Partitions

Worst-Case

1. Edit Locality
2. Cascading Mismatch
3. Duplicated Partitions

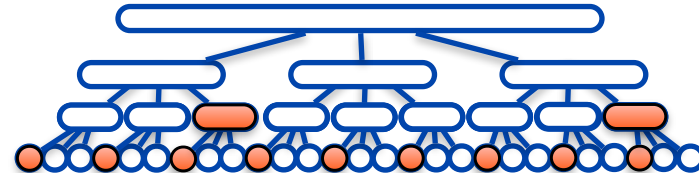


0->A, A->B, B->C

0->A, A->D, D->E ...

Worst-Case

1. Edit Locality
2. Cascading Mismatch
3. Duplicated Partitions

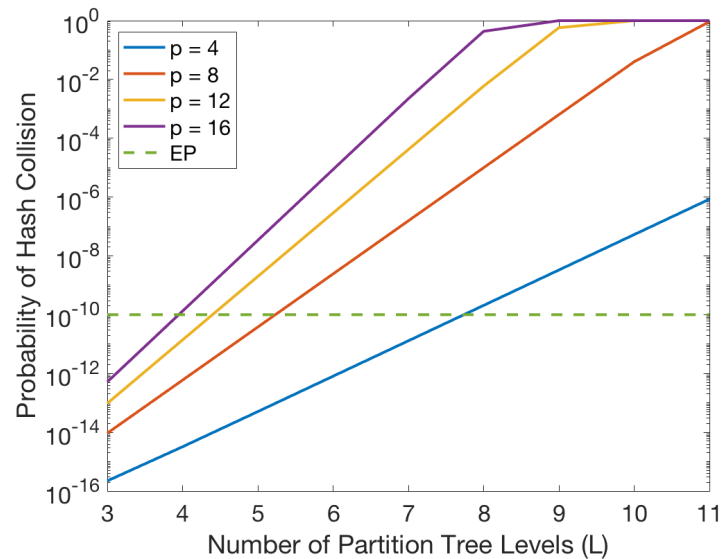


0 $\rightarrow$ A, A $\rightarrow$ B, B $\rightarrow$ C

0 $\rightarrow$ A, A $\rightarrow$ D, D $\rightarrow$ E ...

Failure Modes

1. Hash Collision
2. Set Reconciliation

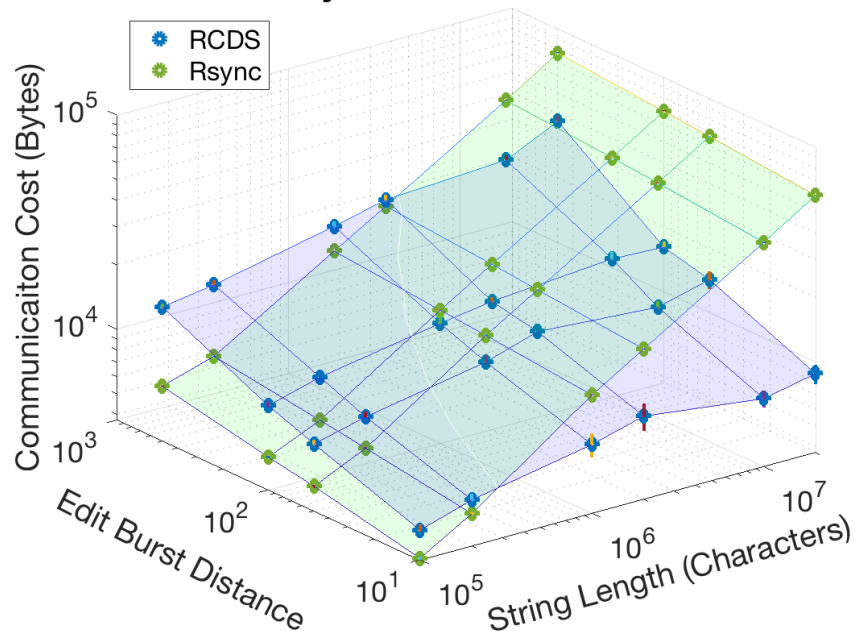


$$Q < 1 - \exp\left(-\frac{(n+1)^2}{2(2^b + 1 - n)}\right)$$

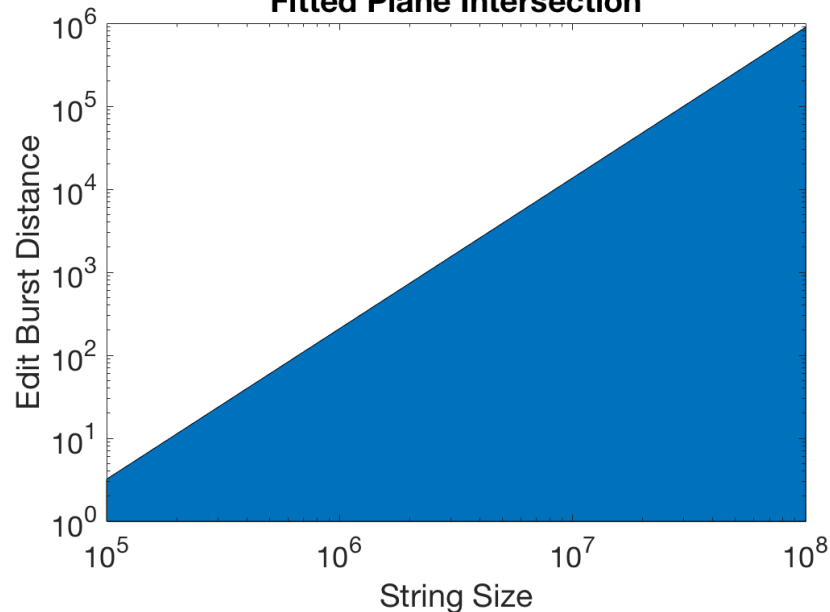
Peyravian, M., On Probabilities of hash values matches, Computer & Security(1998)

RCDS VS. Rsync

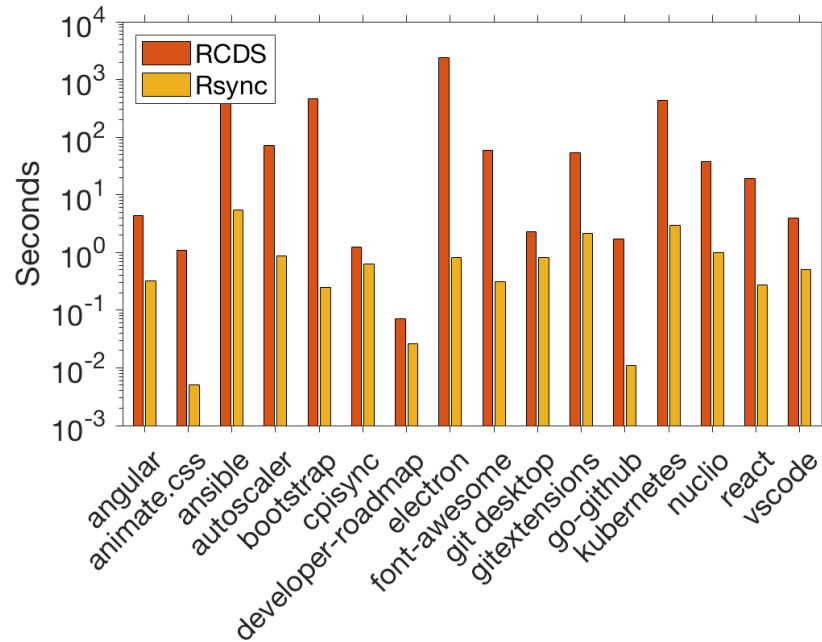
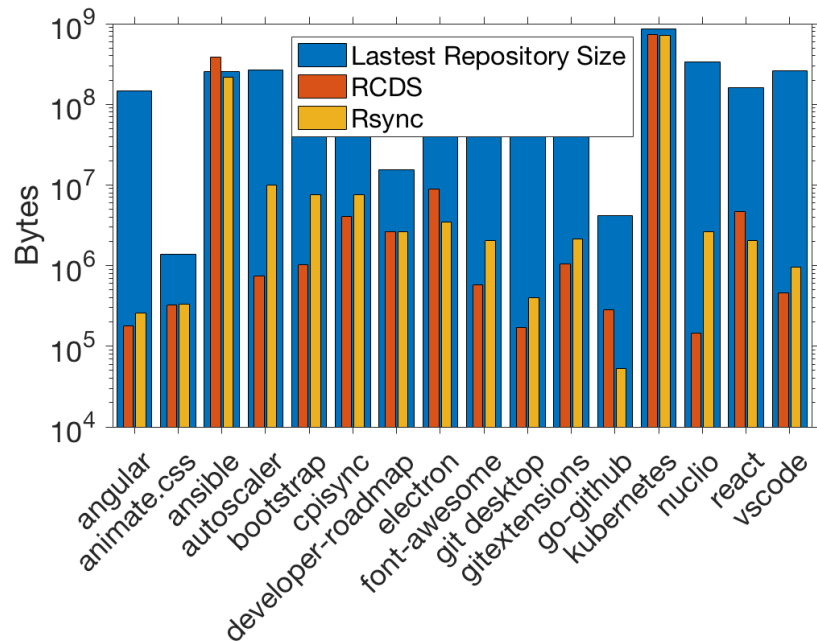
RCDS vs. Rsync Total Communication Cost



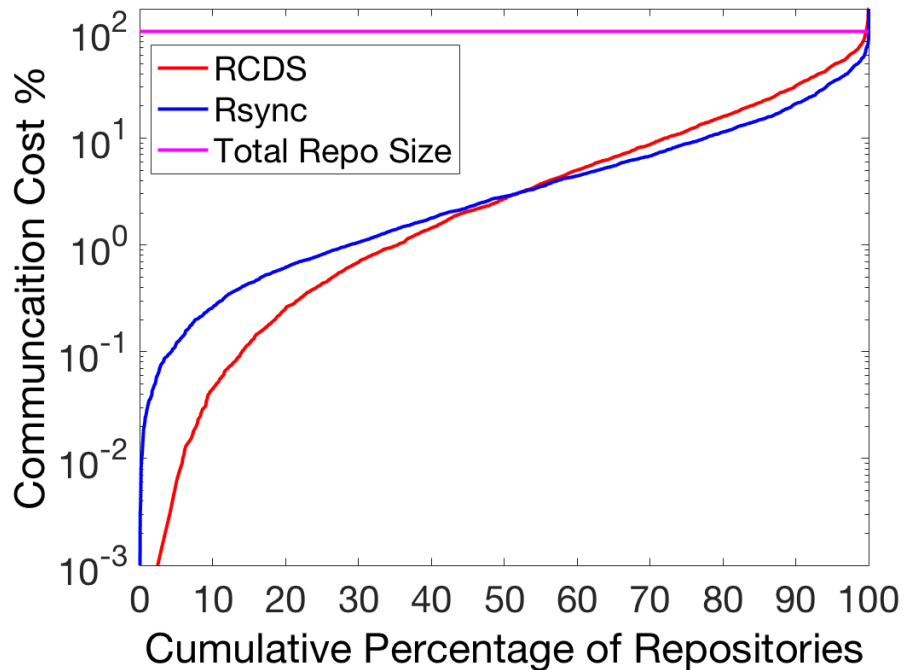
RCDS and Rsync Communication Cost
Fitted Plane Intersection



RCDS VS. Rsync



RCDS VS. Rsync



Conclusion

1. Sub-linear communication cost (empirical)
2. **Scalable** to lengthy string with $O(nL \log(n))$ time
3. Controllable trade-off between computation and communication cost

Future Work

1. ML adjust parameters
2. Hybrid approach
3. Underlying algorithms and data structures
4. Incremental Edits

Thank you

1. Advisor: Prof. Ari Trachtenberg
2. Prof. Osama Alshaykh, Orran Krieger, David Starobinski (Boston University)
3. Apisak Srihamat (Western Digital)

